

Accelerated Algebra and Analysis: Analysis Part

Angus

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Abstract

This is a simple abstract for the Accelerated Algebra and Analysis notes it is focused on the analysis part.

1 Lecture 1

1.1 Introduction

Why the calculation rules exist.

2 Lecture 2

2.1 Three tools from last week

2.1.1 Completeness

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

2.1.2 Approximation property

If $s = \sup A$, then for every $\epsilon > 0$, there exists an $a \in A$ such that

$$s - \epsilon < a \leq s$$

2.1.3 Archimedean property

For every $r \in \mathbb{R}$ with $r > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < r$$

2.2 Sequences

Definition 2.1

Sequence

A sequence is a function $a: \mathbb{N} \rightarrow \mathbb{R}$, written as $(a_n)_{n \in \mathbb{N}}$.

2.2.1 Boundedness

Definition 2.2

Boundedness

- Bounded Above: $\exists M \in \mathbb{R} \forall n \in \mathbb{N} : a_n \leq M$
- Bounded Below: $\exists L \in \mathbb{R} \forall n \in \mathbb{N} : a_n \geq L$
- Bounded: A sequence is bounded if it is both bounded above and bounded below.

2.2.2 Monotonicity

Definition 2.3

Monotonicity: for every $n \in \mathbb{N}$

- | | |
|---------------------------------------|---------------------------------------|
| a) Non-decreasing: $a_{n+1} \geq a_n$ | b) Non-increasing: $a_{n+1} \leq a_n$ |
| c) Increasing: $a_{n+1} > a_n$ | d) Decreasing: $a_{n+1} < a_n$ |

Monotone: A sequence is monotone if it is either non-decreasing or non-increasing.

2.2.3 Convergence

Definition 2.4

Convergence: $\epsilon - N$ Definition

A sequence $(a_n)_{n \in \mathbb{N}}$ converges to a limit $L \in \mathbb{R}$ if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N} : n \geq N \implies |a_n - L| < \epsilon$$

For all $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \epsilon$.

What convergence means: arbitrarily close to a limit, and eventually staying close to it.

$$a_n \rightarrow L \text{ or } \lim_{n \rightarrow \infty} a_n = L.$$

- The sequence converges if it has a real limit.
- If it has no real limit, it diverges.

Notation

∞ is not a real number, the definition contains only finite indices and real inequalities.

Example

Using the definition of convergence, prove that

$$a_n = \frac{1}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Proof. Let $a_n = \frac{1}{n}$. For all $\epsilon > 0$, we need to find $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - 0| < \epsilon.$$

Clearly,

$$|a_n - 0| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n}.$$

Choose

$$N = \left\lceil \frac{1}{\epsilon} \right\rceil.$$

Then $N > \frac{1}{\epsilon}$, so $\frac{1}{N} < \epsilon$. For all $n \geq N$,

$$\frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Thus $|a_n - 0| < \epsilon$. Hence $a_n \rightarrow 0$ as $n \rightarrow \infty$. □

2.3 Convergence and Limitations

2.3.1 Uniqueness of Limits

Definition 2.5

If a sequence $(a_n)_{n \in \mathbb{N}}$ converges to both a and b , $a, b \in \mathbb{R}$, then $a = b$.

Proof. Suppose $a_n \rightarrow a$ and $a_n \rightarrow b$. Let $\epsilon > 0$. Then there exist $N_1, N_2 \in \mathbb{N}$ such that

$$\forall n \geq N_1, |a_n - a| < \epsilon/2,$$

$$\forall n \geq N_2, |a_n - b| < \epsilon/2.$$

Let $m = \max\{N_1, N_2\}$. For all $n \geq m$,

$$|a - b| \leq |a - a_n| + |a_n - b| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, we have $a = b$. □

2.3.2 Limit laws

Theorem 2.1

Addition, scalar multiplication and multiplication laws

Suppose $a_n \rightarrow a$ and $b_n \rightarrow b$, with $c \in \mathbb{R}$. Then

$$a_n + b_n \rightarrow a + b,$$

$$ca_n \rightarrow ca,$$

$$a_nb_n \rightarrow ab.$$

For the sum, we have the following proof:

Proof. Let $\epsilon > 0$. Since $a_n \rightarrow a$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - a| < \epsilon/2.$$

Since $b_n \rightarrow b$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$|b_n - b| < \epsilon/2.$$

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Hence $a_n + b_n \rightarrow a + b$. □

Theorem 2.2

The quotient law

Suppose $a_n \rightarrow a$ and $b_n \rightarrow b$. If

$$b_n \neq 0 \text{ for every } n \in \mathbb{N}, \quad b \neq 0,$$

then

$$\frac{a_n}{b_n} \rightarrow \frac{a}{b}.$$

Notation

Check both denominator conditions: The quotients must be defined, and the limiting denominator must be nonzero.

2.3.3 The squeeze theorem

Theorem 2.3

The squeeze theorem

Suppose $l_n \leq a_n \leq u_n$ for all $n \in \mathbb{N}$, and

$$l_n \rightarrow a \text{ and } u_n \rightarrow a.$$

Then $a_n \rightarrow a$.

Example

Determine whether the sequence

$$a_n = \frac{\sin n}{n}$$

converges, and justify your answer.

2.3.4 Two Ways to Prove Divergence

Theorem 2.4

Boundedness

Every convergent real sequence is bounded.

But the converse is not true: a bounded sequence need not be convergent.

Example

Consider the sequence

$$a_n = (-1)^n.$$

This sequence is bounded since $|a_n| \leq 1$ for all n , but it does not converge because it oscillates between -1 and 1 .

Definition 2.6

Subsequences

A subsequence is

$$(a_{n_k})_{k \in \mathbb{N}}$$

where $n_1 < n_2 < n_3 < \dots$ are natural numbers.

Theorem 2.5

Subsequence test

If $a_n \rightarrow a$, then every subsequence converges to a . Thus two subsequences with different limits imply divergence.

Example

Consider the sequence

$$a_n = (-1)^n$$

and the subsequences

$$\begin{aligned} a_{2k} &= 1, \\ a_{2k-1} &= -1. \end{aligned}$$

This demonstrates the divergence of the original sequence.

2.4 Monotone Convergence and Bolzano-Weierstrass

2.4.1 Monotone Convergence

Theorem 2.6

Monotone Convergence Theorem

- If (a_n) is non-decreasing and bounded above, it converges to

$$\sup\{a_n : n \in \mathbb{N}\}.$$

- If (a_n) is non-increasing and bounded below, it converges to

$$\inf\{a_n : n \in \mathbb{N}\}.$$

The theorem provides the existence of a limit before its value is known. We prove the first part.

Proof. Assume (a_n) is non-decreasing and bounded above.
If $a_1 \leq a_2 \leq a_3 \leq \dots \leq M$, then

$$a_n \rightarrow a = \sup \underbrace{\{a_1, a_2, a_3, \dots\}}_A.$$

For every $\epsilon > 0$, there exists $x \in A$ such that

$$a - \epsilon < x \leq a.$$

Replace $\exists x \in A$ with $\exists n \in \mathbb{N}$ such that

$$x = a_n.$$

Then

$$a - \epsilon < a_n \leq a_{n+1} \leq a_{n+2} \leq \dots \leq a.$$

So for $k = n, n+1, n+2, \dots$, we have

$$|a_k - a| < \epsilon.$$

Take $N = n$. Then for all $k \geq N$, $|a_k - a| < \epsilon$, which proves $a_n \rightarrow a$. $\square$

2.4.2 Bolzano-Weierstrass Theorem

Theorem 2.7

Every bounded sequence of real numbers has a convergent subsequence.

Lemma 2.8

Lemma used in the proof

Every real sequence has a monotone subsequence.

The full sequence need not converge.

Example

Apply Bolzano-Weierstrass Theorem to each sequence

$$a_n = (-1)^n, \quad b_n = \sin n, \quad c_n = n.$$

2.5 Cauchy Sequences and Divergence

2.5.1 Cauchy sequences

Definition 2.7

A real sequence (a_n) is Cauchy if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N: |a_m - a_n| < \epsilon.$$

The definition compares late terms with one another, without specifying a limit.

Cauchy criterion in $\mathbb{R}$ A real sequence is convergent if and only if it is Cauchy.

2.5.2 Divergence to infinity

Definition 2.8

Divergence to infinity

$$a_n \rightarrow +\infty \iff \forall M > 0, \exists N \in \mathbb{N}, \forall n \geq N: a_n > M,$$

$$a_n \rightarrow -\infty \iff \forall M < 0, \exists N \in \mathbb{N}, \forall n \geq N: a_n < M.$$

These are types of divergence.

2.5.3 Useful elementary limits

Standard limits

For $p > 0$ and $r \in \mathbb{R}$ with $|r| < 1$,

$$\frac{1}{n^p} \rightarrow 0, \quad r^n \rightarrow 0.$$

Theorem 2.9

Order and limits

$$a_n \leq b_n \text{ for all sufficiently large } n, \quad a_n \rightarrow a, \quad b_n \rightarrow b \implies a \leq b.$$

3 Lecture 3

3.1 Series

3.1.1 Definition of series

Difference between series and sequences: A series is the sum of the terms of a sequence.

For sequences, convergence is defined as follows.

Definition 3.1

$(a_n)_{n \in \mathbb{N}}$ is convergent to a if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N: |a_n - a| < \epsilon.$$

For series, convergence is defined in terms of the convergence of the sequence of partial sums.

Definition 3.2

We say

$$S \in \mathbb{R}$$

is the sum of the series

$$\sum_{n=1}^{\infty} a_n$$

if the sequence of partial sums $S_m \rightarrow S$ (i.e., $S = \lim_{m \rightarrow \infty} S_m$).

Definition 3.3

The series

$$S = \sum_{n=1}^{\infty} a_n$$

is convergent if the sequence of partial sums

$$S_m = \sum_{n=1}^m a_n = a_1 + a_2 + \cdots + a_m$$

is convergent as $m \rightarrow \infty$.

If the sequence of partial sums

$$(S_m)_{m \in \mathbb{N}}$$

has no real limit, the series is said to be divergent.

Example

For

$$a_k = 2^{-k}, \quad k \in \mathbb{N},$$

the sequence is

$$a_1, a_2, a_3, \dots = 2^{-1}, 2^{-2}, 2^{-3}, \dots$$

Partial sums:

$$\begin{aligned} S_1 &= a_1, & S_2 &= a_1 + a_2, & S_3 &= a_1 + a_2 + a_3, \dots \\ S_m &= 1 - 2^{-m} \rightarrow 1. \end{aligned}$$

Therefore,

$$\sum_{k=1}^{\infty} a_k = 1.$$

Example

$$a_k = 1, \quad a_k = (-1)^k, \quad k \in \mathbb{N}$$

$$\sum_{k=1}^{\infty} 1 = 1 + 1 + 1 + \dots \text{ diverges,}$$
$$\sum_{k=1}^{\infty} (-1)^k = -1 + 1 - 1 + 1 - \dots \text{ diverges.}$$

3.1.2 Geometric series

Definition 3.4

A series with terms cr^k is geometric.

Theorem 3.1

The geometric series

$$\sum_{k=0}^{\infty} cr^k$$

converges if $|r| < 1$ and its sum is

$$\sum_{k=0}^{\infty} cr^k = \frac{c}{1-r}.$$

If $|r| \geq 1$, the series diverges.

Proof.

$$\begin{aligned} S_m &= \sum_{k=0}^m cr^k = c + cr + cr^2 + \dots + cr^m \\ rS_m &= cr + cr^2 + \dots + cr^{m+1} \\ S_m - rS_m &= c - cr^{m+1} \\ S_m(1-r) &= c(1-r^{m+1}) \\ S_m &= \frac{c(1-r^{m+1})}{1-r} \end{aligned}$$

For $|r| < 1$,

$$\sum_{k=0}^{\infty} cr^k = \frac{c(1-r^{m+1})}{1-r} \rightarrow \frac{c}{1-r}.$$

For $|r| = 1$,

$$\sum_{k=0}^{\infty} 1 = 1 + 1 + 1 + \dots \text{ diverges,}$$
$$\sum_{k=0}^{\infty} (-1)^k = 1 - 1 + 1 - 1 + \dots \text{ diverges.}$$

For $|r| > 1$, assume $S_n \rightarrow S$. Then

$$\begin{aligned} r^{m+1} &= S_{m+1} - S_m \\ &\rightarrow S - S = 0, \end{aligned}$$

which is false since $r^{m+1} \not\rightarrow 0$. □

3.1.3 Telescoping series

Definition 3.5

A series is telescoping if its terms can be written as

$$a_k = b_k - b_{k+1}, \quad k \in \mathbb{N},$$

for some sequence $(b_k)_{k \in \mathbb{N}}$.

Remark

$$\begin{aligned} S_m &= \sum_{k=1}^m a_k \\ &= \sum_{k=1}^m (b_k - b_{k+1}) = b_1 - b_{m+1}. \end{aligned}$$

$$S_m \text{ converges} \Leftrightarrow (b_n) \text{ converges.}$$

3.2 Convergence Tests

Theorem 3.2

If

$$\sum_{k=1}^{\infty} a_k \text{ converges, then } a_k \rightarrow 0.$$

Proof.

$$\begin{aligned} 0 &= S - S \\ &= S_{m+1} - S_m \\ &= a_{m+1}. \end{aligned}$$

□

But if $a_k \rightarrow 0$, does not mean the series converges.

For example, the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ has terms tending to 0, but the series diverges.

Theorem 3.3

The harmonic series

$$\sum_{k=1}^{\infty} \frac{1}{k}$$

diverges.

But its terms tend to 0.

Proof. Consider the partial sums

$$S_m = \sum_{k=1}^m \frac{1}{k}.$$

We have

$$S_{2^n} = 1 + \underbrace{\frac{1}{2}}_{\geq \frac{1}{2}} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq \frac{1}{2}} + \underbrace{\frac{1}{5} + \cdots + \frac{1}{8}}_{\geq \frac{1}{2}} + \cdots + \underbrace{\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}}_{\geq \frac{1}{2}}.$$

Each group contributes at least $\frac{1}{2}$, so $S_{2^n} \rightarrow \infty$ as $n \rightarrow \infty$. Hence, the harmonic series diverges. $\square$

Theorem 3.4

The boundedness test for non-negative series

Suppose $a_k \geq 0$ for all $k \in \mathbb{N}$ and $s_n = \sum_{k=1}^n a_k$. Then

$$\sum_{k=1}^{\infty} a_k \text{ converges} \Leftrightarrow (s_n) \text{ is bounded.}$$

Proof. Claim: $(s_n)_{n \in \mathbb{N}}$ is monotone nondecreasing.

$$s_{m+1} - s_m = a_{m+1} \geq 0 \implies s_{m+1} \geq s_m.$$

Therefore, if (s_n) is bounded, it must converge, and hence the series converges. Conversely, if the series converges, the partial sums (s_n) are bounded. $\square$

Theorem 3.5

The comparison test

Suppose $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$. Then

$$\sum_{k=1}^{\infty} b_k \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Proof. Let $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$, and suppose $\sum_{k=1}^{\infty} b_k$ converges. Then there exists $M \in \mathbb{R}$ such that

$$\sum_{k=1}^n b_k \leq M \quad \text{for all } n \in \mathbb{N}.$$

Since $0 \leq a_k \leq b_k$, we have

$$\sum_{k=1}^n a_k \leq \sum_{k=1}^n b_k \leq M \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $\sum_{k=1}^{\infty} a_k$ is bounded and hence converges. $\square$

Example

Consider the series

$$\sum_{k=1}^{\infty} \frac{k+1}{k^3+2}.$$

We have

$$0 \leq \frac{k+1}{k^3+2} \leq \frac{2k}{k^3} = \frac{2}{k^2} \leq \frac{2}{k(k-1)}.$$

Also,

$$\sum_{k=2}^{\infty} \frac{2}{k(k-1)} = 2 < \infty,$$

so

$$\sum_{k=1}^{\infty} \frac{k+1}{k^3+2} \text{ converges.}$$

Theorem 3.6

Suppose a_k is non-negative and non-increasing. Then the series $\sum_{k=1}^{\infty} a_k$ converges if and only if the series $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges:

$$\sum_{k=1}^{\infty} a_k \text{ converges} \Leftrightarrow \sum_{k=1}^{\infty} 2^k a_{2^k} \text{ converges.}$$

The condensed terms sample the original series at indices that are powers of 2.

Proof.

$$\begin{aligned}
S_m &= \sum_{k=1}^m a_k, \\
S_{2^m} &= a_1 + a_2 + \underbrace{(a_3 + a_4)}_{\geq 2a_4} + \underbrace{(a_5 + \cdots + a_8)}_{\geq 4a_8} + \cdots + \underbrace{(a_{2^{m-1}+1} + \cdots + a_{2^m})}_{\geq 2^{m-1}a_{2^m}}, \\
2M &\geq 2S_{2^m} \geq 2a_1 + 2a_2 + 2^2a_{2^2} + \cdots + 2^ma_{2^m} \geq a_1 + \sum_{j=1}^m 2^j a_{2^j}.
\end{aligned}$$

□

Theorem 3.7

p test

For $p \in \mathbb{R}$, the series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges $\iff p > 1$.

Proof. Let

$$a_k = \frac{1}{k^p}.$$

- If $p \leq 0$, then $\frac{1}{k^p} \geq 1$, so the terms do not tend to zero and the series diverges.

- If $p > 0$, then $a_k = \frac{1}{k^p}$ is non-negative and non-increasing, so the condensation test applies:

$$\sum_{j=1}^{\infty} 2^j a_{2^j} = \sum_{j=1}^{\infty} 2^j \frac{1}{(2^j)^p} = \sum_{j=1}^{\infty} (2^{1-p})^j.$$

This geometric series converges exactly when $2^{1-p} < 1$, which holds if and only if $p > 1$.

Therefore, by the condensation test, the original series converges if and only if $p > 1$.

□

Theorem 3.8

The ratio test

Let $a_k \geq 0$ for all $k \in \mathbb{N}$ and suppose

$$r = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k}.$$

Then

- if $r < 1$, $\sum_{k=1}^{\infty} a_k$ converges;
- if $r > 1$, $\sum_{k=1}^{\infty} a_k$ diverges;
- if $r = 1$, the test is inconclusive.

Proof. If $r < 1$, choose s such that $r < s < 1$. Then, for all $k > N$,

$$\frac{a_{k+1}}{a_k} < s.$$

Then for all $k > N$,

$$0 \leq a_{N+k} \leq s^k a_N.$$

The right-hand side is a geometric series with ratio $s < 1$, which converges.

$$\sum_{k=1}^{\infty} a_{N+k} \leq \sum_{k=1}^{\infty} s^k a_N < \infty.$$

So

$$\sum_{k=1}^{\infty} a_{N+k}$$

is convergent. Therefore, the original series $\sum_{k=1}^{\infty} a_k$ also converges.

□

Example

Determine whether each series converges or diverges.

$$\sum_{k=1}^{\infty} \frac{k}{k^2}, \quad \sum_{k=1}^{\infty} \frac{k!}{k^k}.$$

Theorem 3.9

The alternating series test

Suppose (b_k) satisfies

- $b_k \geq 0$ for all $k \in \mathbb{N}$,
- b_k is non-increasing, i.e., $b_{k+1} \leq b_k$ for all k ,
- $b_k \rightarrow 0$.

Then the alternating series

$$\sum_{k=1}^{\infty} (-1)^k b_k$$

converges.

Example

Consider the series

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k}.$$

Definition 3.6

A real series $\sum a_k$ converges absolutely if

$$\sum_{k=1}^{\infty} |a_k| \text{ converges.}$$

It converges conditionally if

$$\sum_{k=1}^{\infty} |a_k| \text{ diverges but } \sum_{k=1}^{\infty} a_k \text{ converges.}$$

To claim conditional convergence, both assertions must be established.

Theorem 3.10

Every absolutely convergent real series is also convergent, i.e.,

$$\sum_{k=1}^{\infty} |a_k| \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Theorem 3.11

The absolute convergence test

Every absolutely convergent real series is convergent:

$$\sum_{k=1}^{\infty} |a_k| \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Proof. Let $\sum_{k=1}^{\infty} |a_k|$ converge.

$$0 \leq b_k = a_k + |a_k| \leq 2|a_k|.$$

Since

$$a_k = b_k - |a_k|,$$

we have

$$\sum_{k=1}^{\infty} b_k \text{ converges by the comparison test.}$$

Also,

$$S_m = \sum_{k=1}^m a_k = \sum_{k=1}^m (b_k - |a_k|) = \sum_{k=1}^m b_k - \sum_{k=1}^m |a_k|.$$

Thus, S_m converges as $m \rightarrow \infty$. □

Example

Classify the type of convergence for the series

$$\sum_{k=1}^{\infty} \frac{\sin k}{k^2}, \quad \sum_{k=1}^{\infty} \frac{(-1)^k}{k}.$$

1. Conditionally convergent:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} \text{ converges by the alternating series test, but}$$

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges.}$$

2. Absolutely convergent:

$$0 \leq \left| \frac{\sin k}{k^2} \right| \leq \frac{1}{k^2},$$

and

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \text{ converges by the } p\text{-series test with } p = 2.$$

3.3 Summary of Convergence Tests

What you notice	A useful first check
Terms do not tend to zero	k th-term test
Non-negative algebraic terms	comparison; p -series
Factorials or exponential powers	ratio test
Non-increasing non-negative terms	condensation
Alternating signs	all three alternating-test hypotheses
General signed terms	absolute convergence first

Example

Choose a test and decide convergence:

$$\sum_{k=1}^{\infty} \frac{k}{3^k} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}.$$